CS338 – Theory of Computation Pietro S. Oliveto

Assignment 11

Submission by: **May 19th - 4pm**

1) [0.5 points]
   Show that the following languages are in NP:

   a. 3SAT = $\{<\phi>\ \mid \phi$ is a satisfiable 3CNF formula$\}$. A 3CNF formula is an AND of clauses where each clause is an OR of 3 literals (variables).
   b. VERTEX COVER = $\{<G,k>\mid G$ is an undirected graph that has a k – node vertex cover$\}$. (A *vertex cover* is a subset of nodes such that every edge of the graph G touches one of the nodes).
   c. TSP = $\{<G,W,k>\mid G$ is an undirected graph, with weights $w_{i,j}$ on each edge that has a tour of length at most k that visits every node exactly once$\}$
   d. MAX-CUT=$\{<G,k>\ \mid G$ has a cut of size k or more$\}$. A *cut* in an undirected graph is a separation of the vertices V into two disjoint subsets S and T. The size of the cut is the number of edges that have one endpoint in S and one in T.
   e. 3COLOR=$\{<G>\ \mid G$ is colourable with 3 colors$\}$. A *colouring* of a graph is an assignment of colours to the nodes such that no node is adjacent to another node with the same colour.

2) [0.5 points]
   Show that NP is closed under union, concatenation and Star.

3) [0.5 points]
   Say whether the following are true or not and argue why:
   a. $P \subseteq NP$
   b. P is closed under complementation i.e., if $A \in P$ then $\bar{A} \in P$
   c. $\overline{SAT} \in NP$
   d. $P \in NP \cap coNP$
   e. If $P = NP$ then $NP = coNP$

4) [1 point]
   Show that 2COLOR=$\{<G>\ \mid G$ is colourable with 2 colors$\}$ is in NP and in P.

5) [BONUS QUESTION – 1 point]
Let $2SAT = \{< \phi > \mid \phi$ is a satisfiable 2CNF formula$\}$. Show that this language is in NP and in P.